STAFF REVIEW · H446 312 source models

Separate from the eight-page pupil workbook. This copy is in a public repository and is NOT access controlled. Check technical correctness before use.

Printed question mapping: Booklet Q1 = original online Q1; Booklet Q2 = original online Q2; Booklet Q3 = original online Q3

Do Now — source answers

1. Standardised mantissa maximising precision.

2. The first two bits differ.

3. NOT A OR NOT B.

4. B∧(A∨¬A)=B.

I Do — source worked steps

1. Predict first — Left-shift mantissa once and lower exponent by one.

2. Explain and check — Explain the reason for the result.

We Do — guided CHECK

1. Predict first
Invert all eight bits then add one, yielding 11110101.
Explain and check
Explain the reason for the result.

You Do — source mark schemes

1. Mantissa0.10100 [2]; exponent3 [2]; value .625×8=5 [2].

2. +23=00010111 [2]; invert11101000 [2]; add1→11101001 [2].

3. (A∨¬A)∧(A∨B) [2]; 1∧(A∨B) [1]; A∨B [1]; substitution gives1 [2].

Exit Ticket — source answer

1. False. Exponent compensates for shifting.